

Quiz 4

③

- (a) apriori probability
- (b) Subjective probability
- (c) apriori probability
- (d) Empirical probability

⑤ $n = 500$

Enjoy shopping for clothing

	Yes	No.	Total
Male	136	104	240
Female	224	36	260
Total	360	140	500

$$(a) P(\text{Enjoy shopping}) = \frac{360}{500} = 0.72$$

$$(b) P(\text{Female and enjoy shopping}) = \frac{224}{500} = 0.448$$

$$(c) P(\text{Female or enjoy shopping}) = \frac{260}{500} + \frac{360}{500} - \frac{224}{500} = \frac{396}{500} = 0.792$$

$$(d) P(\text{Female or Male}) = \frac{240}{500} + \frac{260}{500} - \frac{0}{500} = 1$$

(7)

	B	B'	tot
A	10	30	40
A'	25	35	60
tot	35	65	100

$$(a) P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{10/100}{35/100} = \frac{10}{35} = \underline{0.2857} \rightarrow$$

$$(b) P(A'|B') = \frac{P(A' \cap B')}{P(B')} = \frac{35/100}{65/100} = \frac{35}{65} = \underline{0.5385} \rightarrow$$

$$(c) P(A|B') = \frac{P(A \cap B')}{P(B')} = \frac{30/100}{65/100} = \frac{30}{65} = \underline{0.4615} \rightarrow$$

(d) Statistical independence

$$P(A|B) = P(A) \text{ and } P(B|A) = P(B)$$

$$\text{Now: } P(A|B) = 0.2857$$

$$P(A) = \frac{40}{100} = 0.4$$

Thus, $P(A|B) \neq P(A)$. A and B are statistically dependent

(8) Given

$$P(A \cap B) = 0.4 \text{ and } P(B) = 0.8$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.4}{0.8} = 0.5 \rightarrow$$

(10) Given

$$P(A) = 0.3, P(B) = 0.4 \text{ and } P(A \cap B) = 0.2$$

Statistical independence

$$P(A|B) = P(A) \text{ and } P(B|A) = P(B)$$

$$\text{Now: } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.4} = 0.5$$

$$P(A) = 0.3$$

Thus, $P(A|B) \neq P(A)$. A and B are not statistically independent.

(15) 9 gloves

→ 2 left-handed gloves
→ 7 right-handed gloves

(a)

$R_1 \Rightarrow$ first draw is a left-hand glove

$R_2 \Rightarrow$ second draw is a right-hand glove

Without replacement makes R_1 and R_2 dependent events

(4)

Thus,

$$\begin{aligned}
 P(R_1 \cap R_2) &= P(R_2 | R_1) P(R_1) \\
 &= P(R_1) P(R_2 | R_1) \\
 &= \left(\frac{7}{9}\right) \left(\frac{6}{8}\right) = \frac{42}{72} = 0.5833 \rightarrow
 \end{aligned}$$

(b) $R_1 \Rightarrow$ first draw is a right-hand glove
 $R_2 \Rightarrow$ second draw is a right-hand glove

$L_1 \Rightarrow$ first draw is a left-hand glove
 $L_2 \Rightarrow$ second draw is a left-hand glove

Without replacement makes L_i and R_i dependent events

Thus,

$$\begin{aligned}
 P(L \cap R) &= P(L_1 \cap R_2) + P(R_1 \cap L_2) \\
 &= P(L_1) P(R_2 | L_1) + P(R_1) P(L_2 | R_1) \\
 &= \left(\frac{2}{9}\right) \left(\frac{7}{8}\right) + \left(\frac{7}{9}\right) \left(\frac{2}{8}\right) \\
 &= \frac{14}{72} + \frac{14}{72} = \frac{28}{72} = 0.3889 \rightarrow
 \end{aligned}$$

(c)

 $L_1 \Rightarrow$ first draw is a left-hand glove $L_2 \Rightarrow$ second " " " $L_3 \Rightarrow$ third " " "

With replacement makes L_1, L_2 and L_3 independent events

$$\text{Thus } P(L_1 \cap L_2 \cap L_3) = P(L_1) P(L_2) P(L_3)$$

$$= \left(\frac{2}{9}\right) \left(\frac{2}{9}\right) \left(\frac{2}{9}\right)$$

$$= \left(\frac{2}{9}\right)^3 = \underline{0.0109}$$

$$(d) \rightarrow (a) \quad P(R_1 \cap R_2) = P(R_1) P(R_2)$$

$$= \left(\frac{7}{9}\right) \left(\frac{7}{9}\right)$$

$$= \left(\frac{7}{9}\right)^2 = \underline{0.6049}$$

$$\rightarrow (b) \quad P(L \cap R) = P(L_1 \cap R_2) + P(R_1 \cap L_2)$$

$$= P(L_1) P(R_2) + P(R_1) P(L_2)$$

$$= \left(\frac{2}{9}\right) \left(\frac{7}{9}\right) + \left(\frac{7}{9}\right) \left(\frac{2}{9}\right)$$

$$= \frac{28}{81} = \underline{0.3457}$$

(6)

(16) Given

$$P(B) = 0.05, P(A|B) = 0.8, P(B') = 0.95, P(A|B') = 0.4$$

Calculate

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$= \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B')P(B')}$$

$$= \frac{(0.8)(0.05)}{(0.8)(0.05) + (0.4)(0.95)}$$

$$= \frac{0.04}{0.04 + 0.38} = \frac{0.04}{0.42}$$

$$= 0.0952 \rightarrow$$

Note:

	B	B'	tot
A	0.04	0.38	0.42
A'	0.01	0.57	0.58
tot	0.05	0.95	1

(20) Define :

$S_1 \Rightarrow$ huge success

$S_2 \Rightarrow$ modest success

$S_3 \Rightarrow$ break even

$S_4 \Rightarrow$ losers

$R \Rightarrow$ favourable review

Given:

$$P(S_1) = 0.1, P(S_2) = 0.2, P(S_3) = 0.4$$

$$P(S_4) = 0.3$$

$$P(R|S_1) = 0.99, P(R|S_2) = 0.7, P(R|S_3) = 0.4$$

$$P(R|S_4) = 0.2$$

(a)	i	$P(S_i)$	$P(R S_i)$	$P(R S_i)P(S_i)$	$P(S_i R)$
	1	0.1	0.99	0.099	0.2157
	2	0.2	0.7	0.14	0.3050
	3	0.4	0.4	0.16	0.3486
	4	0.3	0.2	0.06	0.1307
				<u>0.459</u>	

$$\begin{aligned} \text{(b)} \quad P(R) &= P(R \cap S_1) + P(R \cap S_2) + P(R \cap S_3) + P(R \cap S_4) \\ &= P(R|S_1)P(S_1) + P(R|S_2)P(S_2) + P(R|S_3)P(S_3) + \\ &\quad P(R|S_4)P(S_4) \end{aligned}$$

$$= 0.099 + 0.14 + 0.16 + 0.06 = \underline{0.459} \rightarrow$$

Note:

	S_1	S_2	S_3	S_4	Total
R	0.099	0.14	0.16	0.06	$P(R) = 0.459$
R'	0.001	0.06	0.24	0.24	$P(R') = 0.541$
Total	$P(S_1) = 0.1$	$P(S_2) = 0.2$	$P(S_3) = 0.4$	$P(S_4) = 0.3$	1

$$P(R \cap S_1) = P(R|S_1)P(S_1) = (0.99)(0.1) = 0.099$$

$$P(R \cap S_2) = P(R|S_2)P(S_2) = (0.7)(0.2) = 0.14$$

$$P(R \cap S_3) = P(R|S_3)P(S_3) = (0.4)(0.4) = 0.16$$

$$P(R \cap S_4) = P(R|S_4)P(S_4) = (0.2)(0.3) = 0.06$$

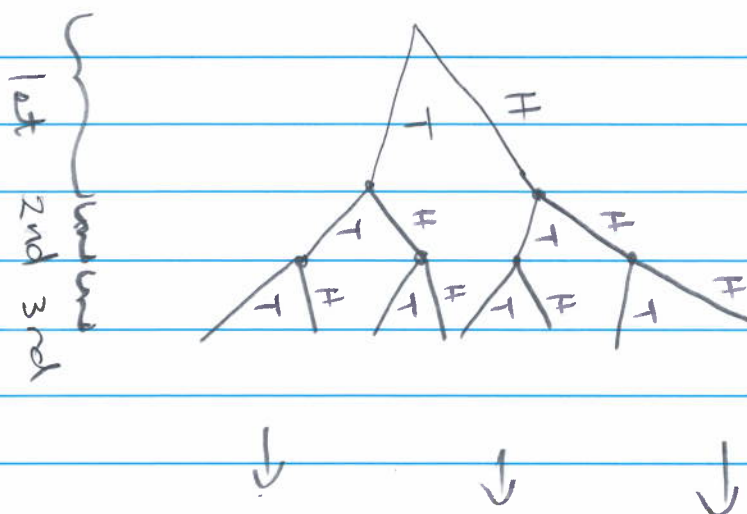
Thus, 45.9% received a favourable review and 54.1% did not receive favourable reviews.

(24)

$n = 7$ times the coin is tossed

$k = 2$ sides of the coin

$$\therefore k^n = 2^7 = 128$$



7th

T, T, T, T, T, T, T

...

H, H, H, H, H, H, H
H, H, H, H, H, H, T
H, H, H, H, H, T, H
H, H, H, H, T, H, H
H, H, H, T, H, H, H
H, H, T, H, H, H, H
H, T, H, H, H, H, H
T, H, H, H, H, H, H

Outcomes

(25)

$n = 7$ times the die is tossed

$k = 6$ sides of the die

$$\therefore k^n = 6^7 = 279936$$

(28)

$\{E, L, O, V\}$

$$4! = 24 \rightarrow$$

$\therefore \{E, L, O, V\}$

$\{E, L, V, O\}$

$\{E, V, O, L\}$

⋮

Outcomes $\Rightarrow 24$

(33)

 $\{B_1, B_2, B_3, B_4, B_5, B_6, B_7\}$
 $n = 7$ Books

 $x = 4$ can fit in bag

Combinations of 4?

$$\begin{aligned}
 {}^n C_x &= \frac{n!}{x!(n-x)!} = \frac{7!}{4!(7-4)!} \\
 &= \frac{5040}{(24)(6)} = \frac{5040}{144} = 35 \rightarrow
 \end{aligned}$$

(34)

 $n = 100$ numbers in total

 $x = 2$ winning numbers

Combinations of 2?

$$\begin{aligned}
 {}^n C_x &= \frac{n!}{x!(n-x)!} = \frac{100!}{2!(100-2)!} \\
 &= \frac{100!}{2! 98!} \\
 &= \frac{(100)(99)(98!)}{2! 98!} \\
 &= 4950 \rightarrow
 \end{aligned}$$

(38)

Given

$$P(A) = 0.3, P(A' \cap B) = 0.22, P(B|A) = 0.6$$

Complete the table

	A	A'	Total
B	0.18	0.22	0.4
B'	0.12	0.48	0.6
Total	0.3	0.7	1

$$(a) P(A \cap B) = P(B|A) P(A) = (0.6)(0.3) = \underline{0.18}$$

$$(b) P(B) = P(A \cap B) + P(A' \cap B) = 0.18 + 0.22 = \underline{0.4}$$

$$(c) P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.18}{0.4} = \underline{0.45}$$

$$(d) P(\text{At least one of } A \text{ or } B) \\ = P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ = 0.3 + 0.4 - 0.18 = \underline{0.52}$$

$$(e) P(\text{Exactly one of } A \text{ or } B) \\ = P(A \cup B) - P(A \cap B)$$

$$= 0.52 - 0.18 = \underline{0.34}$$